Porcentômetro

Completeness Assessment System

Catálogo Taxonômico da Fauna do Brasil (CTFB)

User Manual — English

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Access the application at:

https://comfortable-emotion-production-e366.up.railway.app/

The **Porcentômetro** is a web-based tool developed for the *Catálogo Taxonômico da Fauna do Brasil* (CTFB). It provides two complementary analyses: (1) the **Data Completeness** analysis, which measures how thoroughly the mandatory fields are filled in for species already catalogued, and (2) the **Species-List Completeness** analysis, which compares CTFB records against the Global Biodiversity Information Facility (GBIF) to identify species known from Brazil that are absent from the catalogue.

1. Accessing the System

Open the application in any modern browser:

https://comfortable-emotion-production-e366.up.railway.app/

You will be presented with a login page. Enter your username and password. If this is your first login after being added by an administrator, you will be prompted to create your own password before continuing.

1.1 Authorization

Each user is associated with one or more taxa listed in the researchers spreadsheet (*pesquisadores.xlsx*). You may analyse any taxon within your area of responsibility and all taxonomically subordinate groups. Users assigned to *Animalia* have access to the entire catalogue. Administrators may analyse any taxon.

1.2 First Access — Setting Your Password

Users added by an administrator via the bulk-import tool do **not** have a password assigned yet. On first login, the system detects this and redirects you to a password-creation screen before granting access. Follow the steps below:

1	Open the application URL and type your username in the login form. Leave the password field **empty** and click *Login* (or enter any value — it will be ignored at this stage).
2	The system recognises that no password has been set and automatically redirects you to the **Set Password** screen, which displays a welcome message with your username.
3	Enter a new password of **at least 6 characters** in the "New password" field, then retype it in the "Confirm password" field and click *Set password and continue*.
4	If the two entries match and meet the minimum length, your password is saved and you are immediately logged in and taken to the main analysis page.
5	From this point on, use your username and the password you just created on every subsequent login.

If the passwords do not match or are shorter than 6 characters, an error message is displayed on the same screen and you are asked to try again. No data is lost in this case.

1.3 Forgotten Password

If you forget your password, you cannot reset it yourself. There are two ways to recover access:

- **Contact your administrator** — the administrator can click *Reset password* next to your account in the Admin Dashboard. This clears your current password and sets the "must set password" flag, so the next time you log in you will be taken to the password-creation screen exactly as in the

first-access flow.

- **Write to the support address** — send an e-mail to `faunadobrasilctfb@gmail.com` identifying yourself and your username. The team will ask an administrator to reset your account.

1.3 Main Page

After login the main page shows two analysis-type cards. Click one to select the analysis type, enter a taxon name in the text field (you may click a chip from your authorized taxa list to fill it automatically), then press the action button.

2. Data Completeness Analysis

This analysis evaluates the fraction of mandatory fields that are filled in for each accepted species (*taxonomicstatus* = *NOME_ACEITO*, *taxonrank* = *ESPECIE* or *SUBESPECIE*) within the requested taxon. Results are aggregated by family, order, class, sub-class, and phylum and include the responsible researcher for each group.

2.1 The 15 Evaluated Fields

The following fields are always evaluated unless an ecological exclusion rule applies (see §2.2):

Field	Description
`bibliographicreference`	Primary bibliographic reference for the taxon.
`country`	Country of occurrence (Brazil = BR).
`establishmentmeans`	Native, invasive, introduced, etc.
`endemicbrazil`	Whether the species is endemic to Brazil.
`environment`	Environment type: epicontinental, marinho, or both.
`locality`	Occurrence locality description.
`epicontinentaldomain`	Phytogeographic domain for freshwater/terrestrial species.
`marinedomain`	Marine biogeographic province (sea species only).
`lifeform`	Life-form / feeding guild (herbívoro, parasita, etc.).
`habitat`	Habitat description.
`vernacularname`	Common name in Portuguese.
`vernacularnamelanguage`	Language of the common name.
`vernacularnamelocality`	Region to which the common name applies.
`animalhostnames`	Animal host species (non-herbivores only — see §2.2).
`vegetalhostnames`	Plant host species.

2.2 Ecological Exclusion Rules

Two fields can be excluded from both the numerator and the denominator when they are biologically inapplicable:

Field excluded	Condition	Rationale
`marinedomain`	environment = *epicontinental*	A freshwater/terrestrial species has no marine biogeographic province.
`animalhostnames`	lifeform = *herbívoro*	Herbivores feed on plants; the animal-host field is not applicable.

Example: a freshwater beetle (environment = epicontinental) is evaluated against 14 fields instead of 15 because marinedomain is excluded from both the count of filled fields and the total applicable fields. The completeness percentage is therefore calculated over the applicable denominator only.

2.3 Completeness Formula

For each species the completeness score is computed as:

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completeness (%) = (filled applicable fields) / (total applicable fields) × 100
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The score is then aggregated per taxonomic group using the mean, minimum, and maximum across all species in that group. The **amplitude** (max – min) indicates within-group heterogeneity.

2.4 Researcher Attribution

Each aggregated row lists the researcher (*nome_completo*) responsible for that group. The lookup climbs the taxonomic hierarchy starting at the group's own rank and moving upward until a responsible researcher is found. When the researcher is found at a rank higher than the row's own rank, the cell is annotated with "(via <rank>)" to make the delegation transparent.

2.5 Output Files

Results are packaged in a ZIP file containing:

Sheet / File	Contents
`completeness.xlsx`	Six spreadsheet tabs: by_species, by_family, by_order, by_class, by_subclass, by_phylum. Each tab includes the taxonomic context columns, species count, mean/min/max completeness, amplitude, and responsible researcher.
`by_family.pdf`	Horizontal bar chart of mean completeness by family, sorted alphabetically, with min–max error bars. The most complete family is highlighted in red.
`by_order.pdf`	Same chart aggregated by order.
`by_class.pdf`	Same chart aggregated by class.
`by_subclass.pdf`	Same chart aggregated by subclass (omitted if no subclass data).
`by_phylum.pdf`	Same chart aggregated by phylum (omitted for single-phylum queries).

3. Species-List Completeness (GBIF)

This analysis compares the CTFB accepted species list against species recorded in Brazil on the Global Biodiversity Information Facility (GBIF), identifying species that occur in Brazil but are absent from the catalogue. The analysis runs entirely in the background and reports progress in real time on the status page.

3.1 How It Works

The analysis is divided into four sequential phases:

Phase 1 — Taxon resolution

The taxon name you entered is resolved to a GBIF internal identifier (*usageKey*) via the GBIF Species Match API (api.gbif.org/v1/species/match). An error is returned if the name is not found.

Phase 2 — Species key collection

The GBIF Occurrence Search API is queried with filters country=BR and taxonKey=<resolved key>, using facets on the speciesKey field to retrieve the list of all species with at least one occurrence record in Brazil. Pagination is applied automatically for taxa with more than 50 000 species.

Phase 3 — Name resolution

Each species key is resolved to a canonical binomial name (genus + specific epithet) and its higher taxonomy (family, order, class, phylum) via the GBIF Species API. Up to 20 parallel requests are made simultaneously, matching GBIF's rate limit. For large taxa this phase takes the most time.

Phase 4 — Comparison with CTFB

CTFB species with taxonomicstatus = NOME_ACEITO and taxonrank = ESPECIE are extracted and represented as lowercase canonical binomials (genus + specificepithet). GBIF binomials not found in this set are classified as absent from the catalogue.

3.2 Matching Method and Caveats

Matching is done on the **lowercase canonical binomial** (genus + specificEpithet). This means:

- Species with identical binomials in GBIF and CTFB are counted as present, regardless of author or year.
- Nomenclatural discrepancies — such as genus synonymy, spelling variants, or different taxonomic treatments — will cause a species to appear as "absent" even if it is represented under a different name in CTFB.
- Infraspecific taxa (subspecies, varieties) are excluded from both sides of the comparison.
- Manual review of the gap list is recommended, especially for groups with known nomenclatural instability.

3.3 Output File

A single XLSX file is downloaded with three sheets:

Sheet / File	Contents

`Ausentes_no_CTFB`	All GBIF species (Brazil) absent from CTFB, sorted by phylum → class → order → family → name. Columns: phylum_gbif, classe_gbif, ordem_gbif, familia_gbif, nome_cientifico, gbif_species_key.
`Resumo_por_Familia`	Summary per family: number of GBIF species absent from CTFB, total GBIF species in that family, and percentage absent. Sorted by number of absent species (descending).
`Presentes_no_CTFB`	GBIF species that were matched to a CTFB NOME_ACEITO record. Useful for cross-checking and understanding coverage.

3.4 Expected Run Time

Taxon scale	Approx. GBIF species	Estimated time
Small order (e.g. Nematomorpha)	< 200	10–30 seconds
Medium order (e.g. Araneae)	~5 000	2–5 minutes
Large order (e.g. Coleoptera)	~100 000	15–30 minutes
Full phylum (e.g. Arthropoda)	> 200 000	1–3 hours

The status page auto-refreshes every 2 seconds and shows a progress bar. You may close the tab and return later; the job continues server-side until completion or until the server is redeployed.

4. Administration

Administrator accounts have access to the Admin Dashboard via the link at the bottom of the main page.

4.1 User Management

- **Bulk import** — paste a list of usernames (logins) from *pesquisadores.xlsx*. Each user is created with the flag "must set password on first login". Users not found in the spreadsheet are rejected.
- **Individual add** — create a single user with an immediately usable password.
- **Export CSV** — download a backup of all user accounts including hashed passwords. Recommended before every redeployment when using SQLite.
- **Restore CSV** — re-import users from a previously exported CSV. Existing accounts are skipped; hashed passwords are restored so users do not need to reset their passwords.

4.2 Database Persistence

The system supports two database backends:

Backend	Persistence	How to activate
PostgreSQL	Permanent — survives all redeployments.	Set the DATABASE_URL environment variable to a postgresql:// connection string (e.g. from Railway).
SQLite	Ephemeral — data is lost on each redeployment.	Default when DATABASE_URL is not set. A warning banner is shown on the Admin Dashboard.

4.3 Data Sources

The system loads two Google Sheets on startup. If a Sheet is unreachable (e.g. offline), the system falls back to local XLSX files stored in the *data/* directory of the server.

File	Contents
`pesquisadores.xlsx`	Researcher registry: login, nome_completo, táxons, institution. Drives user authorization and researcher attribution in output tables.
`planilha_extendida.xlsx`	Extended CTFB catalogue with all Darwin Core fields plus manual fields. This is the source for all completeness calculations.

5. Interpreting Results

5.1 Data Completeness Charts

Each PDF chart shows a horizontal bar per taxonomic group. The bar length represents the mean completeness (%). Error bars extend to the minimum and maximum completeness found within the group. The bar with the highest mean is coloured red; all others are blue. Values are annotated to the right of each bar.

A group with 100% mean completeness and zero amplitude is perfectly homogeneous — every species has all applicable fields filled. A wide amplitude indicates that some species are well-documented while others lack many fields, signalling where curatorial effort should be focused.

5.2 Species Gap List

The *Ausentes_no_CTFB* sheet lists candidate species for inclusion in the catalogue. Before acting on an entry, always consider:

- Nomenclatural differences — the species may already be in CTFB under a different (but synonymous) binomial.
- Taxonomic scope — GBIF occurrences include records from all habitats; some species may be marine and outside the scope of CTFB's current coverage.
- Record quality — some GBIF records are misidentified or lack coordinate precision. High-uncertainty records should be verified.
- Subspecific taxa — the analysis covers species level only; subspecies absent from CTFB are not flagged.